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Network Penetration Test

Reconnaissance & Service Enumeration

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Target: 173.2.76.3 (internal lab target)

Start of testing: August 1, 2026

End of testing: August 1, 2026

1. Executive Summary

A network reconnaissance assessment was performed from the exercise's scanning host against the local lab subnet. The objective was to discover live hosts, identify the target by its exposed services, and fingerprint those services with Nmap version detection – specifically to determine the port running SMTP, which was deliberately placed on a non-standard high port.

A single target (173.2.76.3) was found exposing more than twenty open TCP ports. Version detection identified numerous outdated, well-known-vulnerable services (the host is a Metasploitable image), and located SMTP (Postfix smtpd) on a high port that a plain scan reported only as unknown. The exposed attack surface is rated **High**.

Observation	Rating
Excessive exposed attack surface of outdated, known-vulnerable services	High

2. Scope and Objectives

The assessment was limited to reconnaissance of the target below, from the designated scanning host only.

Item	Detail
Scanning host	attack_tools (173.2.76.2), over SSH
Subnet	173.2.76.0/29 (8 addresses)
Target host	173.2.76.3 (crvekrh_meta2_1 – Metasploitable 2)
Tool	Nmap 7.70
Fingerprint parameters	ports 1025-65535, version detection (-sV), fastest timing (-T5)
Objective	Identify live hosts and the port running the SMTP service.

3. Methodology

Testing followed a standard reconnaissance workflow (MITRE ATT&CK T1046, Network Service Discovery):

- **Host identification.** The scanning host's own address was read to determine the subnet.
- **Host discovery.** A scan of the /29 identified the host with multiple open ports.
- **Port enumeration.** The high-port range was scanned to list open ports.
- **Service fingerprinting.** Version detection banner-grabbed each open port to identify the service.

4. Findings

4.1 Host discovery

Scanning the /29 subnet distinguished the target by its footprint – many open ports versus one on the scanning host:

```
sudo nmap -F 173.2.76.0/29

Nmap scan report for crvekruh_meta2_1 (173.2.76.3)
 80/tcp open http      111/tcp open rpcbind
5900/tcp open vnc        6000/tcp open X11      ... (many more)
Nmap scan report for attack_tools (173.2.76.2)
 22/tcp open  ssh      (scanning host - not the target)
```

4.2 Service fingerprinting (the SMTP port)

A plain scan labelled the high ports unknown; version detection resolved them to real services. SMTP was found on a non-standard high port, identified by its banner:

```
sudo nmap -p 1025-65535 -sV -T5 173.2.76.3

PORT      STATE SERVICE VERSION
<redacted> open  smtp   Postfix smtpd    <-- exercise answer (port redacted)
```

The SMTP port number is the exercise deliverable (redacted in this report; recorded in the submission). Because SMTP was moved off its standard port 25, only version detection – reading the Postfix smtpd banner – could identify it.

4.3 Exposed attack surface

Version detection revealed that the target is a Metasploitable image exposing many outdated services, several with well-known remote-code-execution vulnerabilities:

Service (from -sV)	Known issue
vsftpd 2.3.4	Backdoor – CVE-2011-2523
UnrealIRCd	Backdoor – CVE-2010-2075
distccd v1	Remote code execution – CVE-2004-2687
Samba 3.x	usermap_script RCE – CVE-2007-2447
Others	MySQL, PostgreSQL, Tomcat, telnet, VNC, X11, rpcbind, ...

For this exercise only the SMTP port is required, but the same scan maps a dozen viable footholds – illustrating why reconnaissance is the foundation of an engagement.

4.4 Impact

A large, outdated exposed surface materially increases risk:

- Each exposed service is a potential entry point; several here have public RCE exploits.
- Non-standard ports provide no protection – version detection identifies services regardless of port.

- An attacker gains a detailed service/version inventory from a single scan, accelerating exploitation.

4.5 Remediation

- **Reduce the attack surface.** Close or firewall unused ports; expose only required services behind a default-deny firewall.
- **Patch or replace outdated software.** Nearly every identified service has a known CVE.
- **Do not rely on non-standard ports** as a security control; obscurity is defeated by version detection.
- **Segment and monitor.** Isolate services and alert on scanning/enumeration behaviour.

5. Conclusion

Standard Nmap reconnaissance discovered the target by its open-port footprint and, using version detection over the high-port range, identified SMTP (Postfix smtpd) on a non-standard high port – the exercise deliverable. The scan also revealed a Metasploitable host exposing numerous outdated, known-vulnerable services. The exposed attack surface is rated High; it should be reduced by closing unused ports, patching services, and not relying on port obscurity.

Appendix A – Command Log

```
sudo ifconfig
sudo nmap -F 173.2.76.0/29
sudo nmap -p 1025-65535 -T4 173.2.76.3      # fast open-port enumeration
sudo nmap -p <open-ports> -sV 173.2.76.3   # version-detect only open ports
sudo nmap -p 1025-65535 -sV -T5 173.2.76.3 # exercise-spec single command
```

Produced for an authorised security training exercise against an intentionally vulnerable lab target, from the exercise's designated scanning host. Confidential.